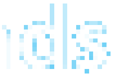
LA HW

Included exercises: 1-2, 16-17

Exercises 1-8

In Exercises 1–8, find the equation of the least-squares line for the given data.

Original Exercise 1



Original Exercise 2

$$(1, 14), (3, 17), (5, 19), (7, 20)$$

$$(1, 30), (2, 27), (4, 21), (7, 14)$$

Exercises 16-19

In Exercises 16–19, an inconsistent system of linear equations $A\mathbf{x} = \mathbf{b}$ *is given. Use the method of least squares to obtain the vectors* $\mathbf{z}$ *for which* $\|A\mathbf{z} - \mathbf{b}\|$ *is a minimum.*

Original Exercise 16

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 3 & 1 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 3 \\ 5 \\ 4 \end{bmatrix}$$

Original Exercise 17

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & -1 & 2 \\ 2 & 1 & 1 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$